

## Genomics in Multiple Myeloma

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### Abstract

Multiple myeloma (MM) is a complex disease that is driven by numerous genetic and epigenetic alterations. Comprehensive oncogenomic analysis indicates the presence of many highly recurrent and highly focal amplifications and/or deletions in the MM genome. Integrated oncogenomic analyses of human MM have identified candidates resident within regions of amplification and/or deletions that are predicted to be involved in MM pathogenesis and progression. The biological behavior and clinical outcome in MM are dependent on these molecular determinants, which are also attractive therapeutic targets. The data obtained from extensive analysis of patient samples, with annotated clinical outcomes, are providing insights into molecular mechanisms of disease behavior, helping to develop sensitive prognostic models, identifying novel therapeutic targets, providing the framework for the development of molecularly based therapies, and, eventually, will help in developing individualized therapy to improve outcomes, with reduced toxicity. *Clin Cancer Res*; 17(6); 1234–42. ©2011 AACR.

### Introduction

Multiple myeloma (MM) is characterized by a significant heterogeneity at multiple levels, such as clinical presentation, biologic characteristics, response to treatment, and clinical outcome. The current data support the hypothesis that this heterogeneity is mainly related to molecular characteristics of the tumor clone. Although genetic changes are the hallmark of the cancer cell, in many hematologic neoplasms, these changes are very limited (like in chronic myeloid leukemia and most acute leukemias). In contrast, solid tumors usually present a wide variety of chromosomal and genomic rearrangements. Myeloma is probably in between these 2 extreme genetic landscapes. Indeed, karyotypes in MM are usually complex, with both quantitative (chromosome number) and qualitative (chromosome structure) abnormalities (1–5). However, despite this complexity, several recurrent changes can be observed, including hyperdiploidy (5), loss of chromosome 13 (6–9), and specific translocations like t(11;14) (q13;q32); refs. (10–12), t(4;14)(p16;q32); refs. (13–16),

or t(14;16)(q23;q32); refs. (17, 18). In this review, we examine how the currently available genomic data can help build pathogenetic and prognostic models that could be used for patient management.

### Pathogenesis

Although complex karyotypic recurrent changes have been described in myeloma, the karyotypic oncogenetic classification, mainly based on hyperdiploidy and 14q32 translocations (19), is only partially confirmed by molecular studies. However, myeloma has now been evaluated at the transcriptional level in several studies, using gene expression profiling. In this approach, CD138+–purified malignant plasma cells are used to extract RNA and hybridized on an array to evaluate expression of genes that are representative of the whole transcribed genome. Using unsupervised bioinformatic methods, tumors are then classified according to gene expression profile similarities. One of the first analyses compared gene expression profiling in cohorts of patients with monoclonal gammopathy of undetermined significance (MGUS) and MM, showing sequential genetic changes from normal plasma cells to malignant plasma cells, in the process, providing clues to the molecular basis for malignant transformation and potential therapeutic targets (Fig. 1; ref. 20). The first molecular classification in 2002 clustered genes according to similarities with either MGUS or human myeloma cell lines (21) and identified 4 classes of MM, according to their similarities to MGUS or MM cell line profiles. In 2003, another study showed that the most relevant profiles were related to Ig gene expression (22). More recently, using unsupervised analyses, 3 reports identified subgroups

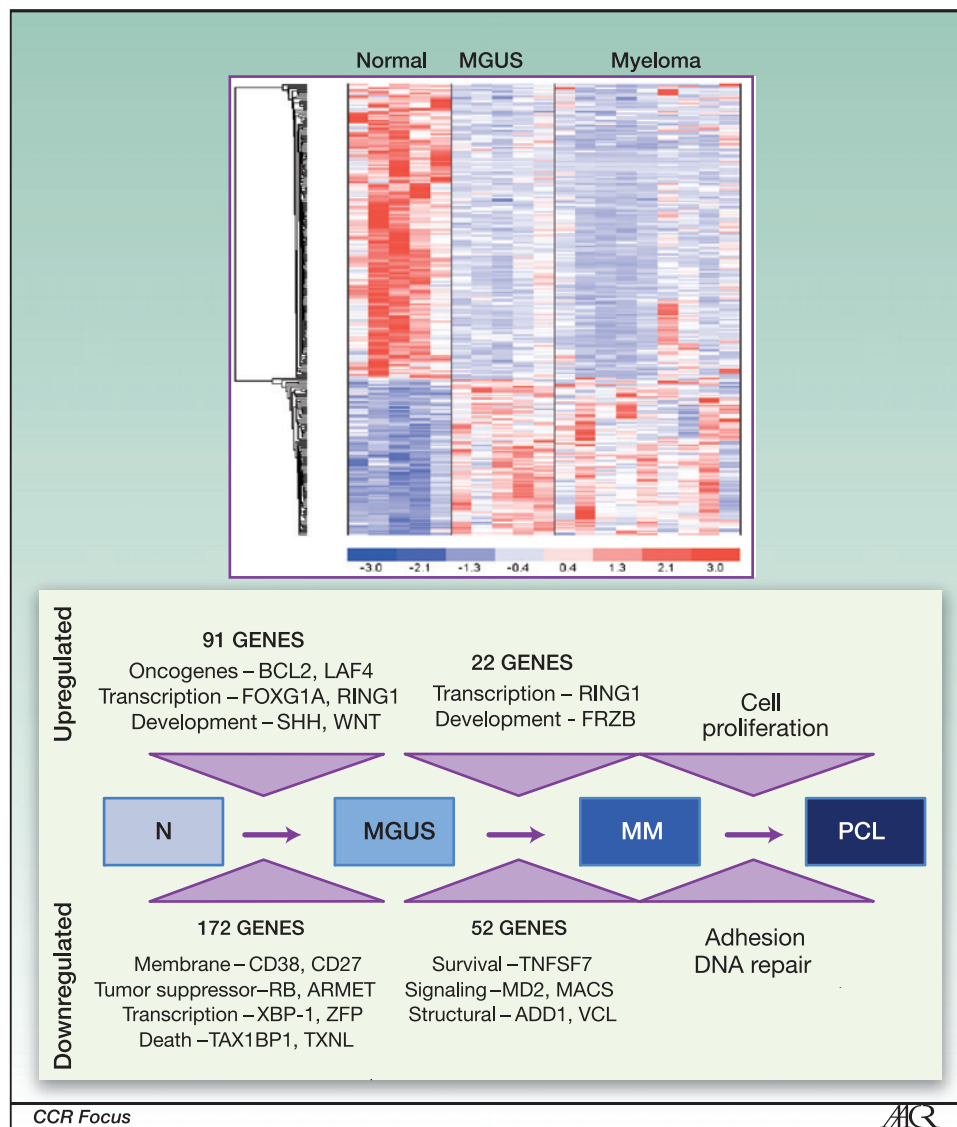
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**Figure 1.** Progression to myeloma: gene expression analysis. High-throughput gene expression profile identifies distinct pattern in myeloma and MGUS compared with normal plasma cells (above). Analysis of the expression profile data identifies a multistep model of progression from normal plasma cells to MGUS cells and to MM. PCL, plasma cell leukemia. Figure adapted with permission from Davies et al. (20).



mostly driven by chromosomal aberrations. The first report identified 8 different subgroups, mainly on the basis of the cyclin D gene expression, and on the different 14q32 recurrent translocations (19). This molecular classification has been refined in 2006, identifying 7 subclasses of myeloma (23). In this pathogenetic model, the first class is defined by the translocation t(4;14) and identified by overexpression of the *MMSET* and/or *FGFR3* genes. The second class is defined by upregulation of 1 of the *MAF* genes, related to the translocations t(14;16) or t(14;20). Cases with *CCND1* or *CCND3* upregulation [because of the translocations t(11;14) or t(6;14)] clustered in 2 different groups, named CD1 and CD2. The CD2 group was characterized by CD20 expression. The fifth group was characterized by hyperdiploidy. The last 2 groups were characterized by a low incidence of bone disease,

according to low dickoff-1 (*DKK-1*) expression, whereas the last group was characterized by high expression of genes involved in proliferation. This molecular classification has been partially confirmed and further refined by a recent study by the HOVON group (24). Although the "low bone disease" group was not confirmed, 3 other subgroups were identified as follows: 1 group enriched by "myeloid" genes (that could be related to plasma cell-sorting problems); 1 group characterized by overexpression of cancer testis antigen genes; and, finally, a group defined by overexpression of positive regulators of the NFκB pathway.

DNA-based techniques, such as array comparative genomic hybridization (25, 26), have identified the role of the NFκB pathway in myeloma. In 2 separate studies, it has been shown that the NFκB pathway can be activated, either

by deletions of NFκB inhibitors (such as TRAF3 or CYLD), or by activation of NFκB activators (such as NIK or CD40). Other studies, based on the analysis of copy number changes by high-density single nucleotide polymorphism (SNP) array, have identified other levels of molecular heterogeneity (27, 28, 29, 30). These studies identified genomic heterogeneity within the hyperdiploid group that was driven by the presence of either chromosome 1q gain and/or chromosome 11 gain, chromosome 13 loss, or chromosome 5 gain, conferring significant outcome difference. Furthermore, these studies showed that the integration of copy number changes and gene expression values allowed the conversion of genomic heterogeneity into the identification of potential cancer target genes. Subclasses of hyperdiploid MM patients with clinical and biological associations were also characterized by gene expression profiling (31). Initial attempts at understanding the genesis of this genomic heterogeneity reside in uncontrolled recombination mechanisms, which may become potential targets in understanding myeloma biology and in developing effective therapeutic strategies (32).

To summarize the tremendous knowledge achieved in the molecular description of myeloma, substantial heterogeneity is identified, which, to date, has prevented the identification of specific definitive entities to be able to classify myeloma into different disease subgroups. However, it is interesting to note that the emerging classifications are essentially based on recurrent chromosomal changes. This situation is highly reminiscent of the non-Hodgkin lymphomas, in which their classification is based on chromosomal changes specific to each entity, such as t(11;14) for mantle cell lymphoma, t(8;14) for Burkitt's lymphoma, or t(14;18) for follicular lymphoma. Further work is needed to confirm this type of classification.

## Prognosis

Although molecular studies have not yet provided a definite myeloma subclassification into specific diseases correlating with biology or clinical behavior, they have contributed to the identification of several prognostic factors that significantly influence patient outcome. Through the use of conventional cytogenetics, recurrent chromosomal changes have been identified and correlated with clinical outcome (Table 1). The abnormalities such as t(4;14), t(14;16), part or whole chromosome 13 deletion, and loss of 17p13 carry a poor prognosis in patients undergoing high-dose therapy, whereas hyperdiploidy and t(11;14) translocations are associated with better outcome. The significance of chromosome 13 deletion remains enigmatic, because it is also observed in patients with MGUS, with an unclear relationship of its transformation to myeloma.

Because myeloma cells have a low proliferative index, the prognostic significance of genetic abnormalities is analyzed by interphase FISH. Among the specific 14q32 translocations discussed previously, t(4;14) is the most important from a clinical point of view. A number of studies have confirmed that patients who display this translocation (15% of patients) have a poor prognosis (33–38). Interestingly, these patients may require a specific therapeutic approach that includes novel agents, such as proteasome inhibitors or immunomodulatory agents. Previously reported del13 is not considered to predict poor outcome by itself. The most important chromosomal change for prognosis is del(17p). Present in 8 to 10% of patients, this deletion is associated with a remarkably short survival, irrespective of the therapy used (37, 39, 40). The molecular target of this deletion could be *TP53*, but no clear biological evidence supports this hypothesis, and mutations are observed in only a subset of patients with del(17p; ref. 41). Finally, several reports have shown that gains of chromosome 1q (observed in one third of the patients) also confer a poor prognosis (42, 43). This abnormality is typically a secondary event, not specific to myeloma, acquired during evolution.

The prognostic significance of molecular changes have been analyzed by high-throughput microarray profiling techniques, focused on copy number alteration (CNA) using either SNP array or gene expression profiling. These techniques are potentially more powerful because they analyze the whole genome.

An analysis of genome-wide CNAs in 192 newly diagnosed uniformly treated MM patients, using high-density SNP arrays, suggested a global genomic instability in MM (28). One of 3 distinct CNA patterns is present in 98% of cases: loss or gain of the chromosome, loss or gain of a whole arm of the chromosome, and/or interstitial losses or gains. Analysis of the most frequent lesions (>10%) have identified 2 main groups: the first group encompasses almost exclusively (with the exception of chromosome 11) either gain or loss of entire chromosomes or interstitial

**Table 1.** Recurrent Cytogenetic Changes in Myeloma

| Type                            | Alterations    | Percent  |
|---------------------------------|----------------|----------|
| Common cytogenetic alterations  | Hyperdiploidy: | 50 to 60 |
|                                 | t(4;14)        | 15       |
|                                 | t(11;14)       | 20       |
|                                 | t(14;16)       | 3        |
|                                 | t(14;20)       | 1        |
|                                 | del; 13 or 13q | 45       |
| Recently identified alterations | del 17p        | 8        |
|                                 | 1q+            | 35       |
|                                 | 1p–            | 30       |
|                                 | 5q+            | 50       |
|                                 | 12p–           | 10       |

gain or loss of the flanking centromeric regions. This group includes gains of chromosomes 3, 5, 7, 9, 11, 15, 18, 19, and 21 and loss of chromosomes 13, 22, and X (in female cases). The second group is characterized by genetic lesions that affect gain or loss of subchromosomal material, including amplification of 1q and 6p, and deletion of 1p, 6q, 8p, 12p, 14q, 16p, 16q, and 20p. The analysis of prognostic significance of CNAs in MM identified amplifications of 1q and deletions of 1p, 12p, 14q, 16q, and 22q to be associated with poor prognosis, whereas amplifications of chromosomes 5, 9, 11, 15, and 19 conferred a superior outcome. A multivariate analysis identified a prognostic model that includes amp(1q23.3), amp(5q31.3), and del(12p13.31) as the most powerful independent adverse markers ( $P < 0.0001$ ), and the prognostic significance of the model has been validated in an independent cohort of 273 patients with MM. These findings, therefore, show the feasibility of molecular karyotyping using SNP profiling to predict outcome in MM. This prognostic model has to be confirmed in independent series. Recurrent cytogenetic changes in myeloma are listed in Table 1 (30).

Two large studies have evaluated the prognostic significance of gene expression profiling to identify poor-risk patient populations. One of the 2 models, the University of Arkansas for Medical Sciences (UAMS) 70-gene model, has 30% of the informative genes mapped to chromosome 1 (44). In the other model, the Intergroupe Francophone du Myélome (IFM) 15-gene model, high-risk patients were enriched in genes controlling proliferation and chromosomal instability, whereas low-risk patients were enriched in hyperdiploid karyotypes (45). It is interesting to note that the 2 models do not have a single common gene, which reflects the redundancy in the genes and pathways that control growth, proliferation, and survival, in addition to differences in the platforms used for the microarray analyses and/or differences in the treatment used to define the patient population. However, in an attempt to validate the techniques, the IFM 15-gene set was shown to be powerful in the UAMS population, but with a lower significance (45). Interestingly, both sets identified patients with a short survival, but none of them identified very good risk patients, probably because of a short follow-up. An international large-scale effort is needed to fully validate a uniform set of genes predictive of outcome, irrespective of the treatment used to make gene expression profiling a routine in clinical practice. Moreover, as CpG methylation affects gene expression and, thus, may be relevant to the pathogenesis and behavior of myeloma cells, a genome-wide methylation profile has been analyzed using microarray. In a recent study, methylation patterns, especially hypomethylation, was capable of distinguishing nonmalignant from malignant plasma cells (46). In fact, differential methylation was also evident at the transition of MGUS cells to MM cells. Interestingly, genes involved in cell-cell signaling and cell adhesion were remethylated in cells from plasma cell leukemia stage, suggesting the

development of independence from the interaction with bone marrow microenvironment cells.

Recently, 2 transcriptome modifiers have been investigated in myeloma. Alternate splicing is an important posttranslational change that alters specificity of gene function. Dysregulated alternative splicing has been reported in myeloma with an effect on overall clinical outcome (47). MicroRNAs (miR) are small noncoding RNA molecules that regulate multiple target genes through cleavage of targeted transcripts and by inducing translational inhibition. Differential expression of the number of miRs has been described in myeloma and MGUS compared with normal plasma cells (48). In 1 study, miRs-21, -106b-25, and -181a/b were overexpressed in MM and MGUS with respect to normal plasma cells, whereas miRs-32 and -17-92, were exclusively overexpressed in MM compared with MGUS. Two target genes of overexpressed miRs, *SOCS-1* and *p300-CBP*, were identified as having an influence on myeloma pathogenesis. Downregulation of miRs 15a/-16 present on chromosome 13 has also been described with potential effect on MM cell proliferation (49); however, its relation with chromosome 13 or 13q34 deletion has not been established (50). Some association between miR expression pattern and molecular and genetic subgroups in myeloma has been described (51, 52); for example, overexpression of miR-let 7e, -125-5p, and -99b located at 19q13.33 in patients with t(4;14) translocation (51) and miRs-1 and -133a in t(14;16) MM has been reported (52). Combined mRNA and miR profiling has identified a miR-mRNA regulatory network with early evidence of differential expression in high-risk disease (53). miRs-192, -194, and -215, downregulated in a subset of MM patients, are correlated with transcriptional activation by p53 and modulation of MDM2 expression, suggesting these miRs as positive regulators of p53 with an important role in MM development (54). Unsupervised clustering analysis of miR expression profile data has also identified groups with different survival outcomes recognizing critical miRs as modulators of gene expression and signaling pathways and has provided potential novel miR and gene targets in MM, to both understand biological behavior and for therapeutic application (55).

### Therapeutic Implications

To date, treatment options have been driven by age and/or physiologic conditions. In patients under 65 years of age, the standard of care is usually a short induction followed by a high-dose melphalan with stem cell rescue. For older patients or those with comorbidities, long-term treatment with a combination is usually chosen. However, the availability and understanding of genomic data have significantly contributed to myeloma therapy. First, both *in vitro* and *in vivo* models have been developed to characterize MM cell-bone marrow stromal cell interactions, as well as signaling pathways controlling growth, survival, drug resistance, or migration within the bone marrow milieu. These studies have delineated the

**Table 2.** Novel Target in Myeloma, Agents, and Stage of Ongoing Clinical Trials

| Target                                         | Agent                     | Clinical study phase | Single agent or combination     |
|------------------------------------------------|---------------------------|----------------------|---------------------------------|
| <b>Cell surface targets</b>                    |                           |                      |                                 |
| FGF3                                           | Dasatinib                 | I-II                 | S                               |
| FGF, PDGF                                      | (mAb)TKI258               | I                    | S                               |
| CD38                                           | mAb                       | I                    | S                               |
| CD40                                           | SGN-40 (mAb)              | I-II                 | S, C (lenalidomide)             |
|                                                | HCD122 (mAb)              | I                    | S                               |
| CD56                                           | huN901-DM1 (C-mAb)        | I                    | S                               |
| CS1                                            | HuLuc63 (mAb)             | II-III               | S, C (lenalidomide, bortezomib) |
| CD138                                          | BT062 (mAb-DM4)           | I                    | S                               |
| RANKL                                          | AMG162 (mAb)              | I-II                 | S                               |
| MUC1                                           | AR20.5 (mAb)              | I-II                 | S                               |
| BAFFR                                          | LY2127399 (mAb)           | I-II                 | S                               |
| CD52                                           | Alemtuzumab (mAb)         | II                   | S                               |
| TRAIL                                          | Apo2L/TRAIL (Apo2 ligand) | I                    | S                               |
|                                                | Mapatumumab               | I-II                 | S                               |
| IGF1/R                                         | IGF1R CP-571 (mAb)        | I                    | S                               |
|                                                | EM164 (mAb)               | I                    | S                               |
| IL-6/R                                         | CNT0328 (mAb)             | II-III               | S, C (bortezomib)               |
|                                                | Altizumab (mAb)           | III                  | S                               |
| VEGF/R                                         | Bevacizumab (mAb)         | II                   | S                               |
|                                                | SU5416                    | II                   | S                               |
|                                                | Zactima (ZD6474)          | II                   | S                               |
| DKK-1                                          | BHQ-880                   | I-II                 | S                               |
| Activin                                        | A                         | I-II                 | S                               |
| KIR                                            | IPH101                    | I-II                 | S                               |
| CXCR3                                          | AMD3100                   | II                   | C (bortezomib)                  |
| <b>Intracytoplasmic and/or nuclear targets</b> |                           |                      |                                 |
| CDK                                            | Alvocidib (NSC649890)     | I                    | S                               |
| CDK and GSK3 $\beta$                           | AT7519M                   | I-II                 | S, C (bortezomib)               |
| IKK                                            | RTA402                    | I                    | S                               |
| Akt                                            | Perofosine                | III                  | C (bortezomib)                  |
| HDAC                                           | Panabino-stat             | III                  | C (bortezomib)                  |
|                                                | Vorinostat                | II-III               | C (bortezomib)                  |
|                                                | Romidopsin                | II-III               | C (bortezomib)                  |
| Farnesyltransferase                            | Tipifarnib (R115777)      | II                   | S, C (bortezomib)               |
| HSP90                                          | KOS953                    | II                   | C (bortezomib)                  |
|                                                | AUY922                    | II                   | C (bortezomib)                  |
|                                                | IPI504                    | I-II                 | C (bortezomib)                  |
| Proteasome                                     | Carfilzomib               | II-III               | S, C (lenalidomide)             |
|                                                | NPI-0052                  | I                    | S                               |
|                                                | MLN9708                   | I                    | S                               |
| Mitochondria                                   | GCS-100                   | I-II                 | C                               |
| mTOR                                           | CCI-779 II                | II                   | C (bortezomib)                  |
|                                                | RAD001                    | II                   | C (lenalidomide, bortezomib)    |
|                                                | INK128                    | II                   | S                               |
| PKC                                            | Enzastaurin               | I-II                 | S, C (bortezomib)               |
| Telomerase                                     | GRN163L                   | I-II                 | S, C (bortezomib)               |

Abbreviations: mAb, monoclonal antibody; S, single agent; C, combination; FGF3, fibroblast growth factor 3; PDGF, platelet derived growth factor; RANKL, RANK ligand; TRAIL, tumor necrosis factor-related apoptosis-inducing ligand; IGF1, insulin like growth factor I; IL-6, interleukin-6; CDK, cyclin dependent kinase.

Data obtained from the following websites: National Cancer Institute ClinicalTrials.gov (<http://clinicaltrials.gov/>), Multiple Myeloma Research Foundation (<http://www.themmr.org/>), and the International Myeloma Foundation (<http://myeloma.org/>).



signaling cascades and molecular mechanisms and identified that MM cell growth is mediated via extracellular signal regulated kinase (ERK)/mitogen activated protein kinase (MAPK), survival via Janus activated kinase (JAK)/STAT, drug resistance via phosphoinositide 3-kinase (PI3K)/Akt, and migration via PKC-dependent signaling cascades. These systems have been used to identify potential novel therapeutic targets, and to validate novel targeted therapies. Genomic studies, especially gene expression profiling, confirmed the significance of these targets, as well as the role and effects of agents directed at these targets. These studies led to the development and U. S. Food and Drug Administration (FDA) approval of bortezomib, a proteasome inhibitor, thalidomide, and lenalidomide, an immunomodulator in MM. Other representative novel agents include the newer proteasome inhibitors carfilzomib and NPI-0052; multitargeted kinase inhibitor PKC-412; histone deacetylase (HDAC) inhibitors; hsp90 inhibitor, telomerase inhibitors; small molecule inhibitors against Akt (perifosine); cyclin-dependent kinase inhibitors; and anti-CD40, anti-CD56, and anti-CD138. The promising targets, agents, and stage of their clinical development is listed in Table 2.

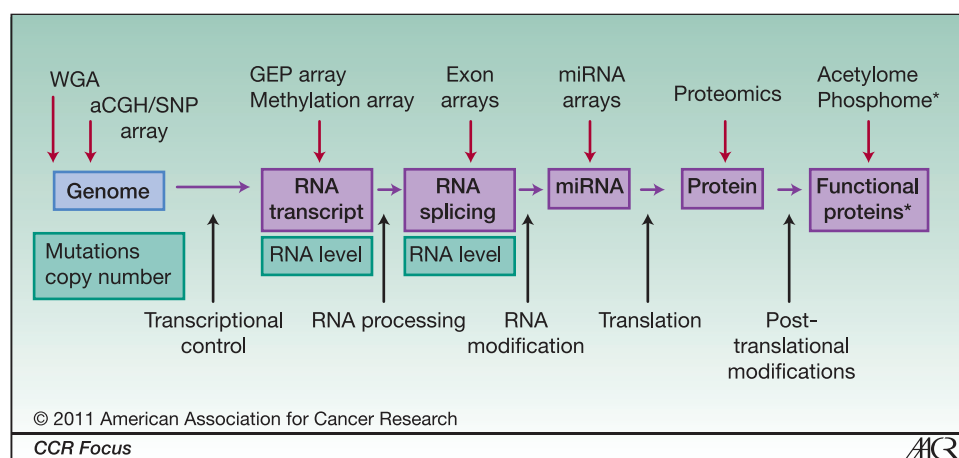
Secondly, the genomic studies have begun to identify new targets, for example, DKK-1, which have been validated *in vitro* and *in vivo*, with translation to clinical studies. Thirdly, these studies have informed the development of combination therapies on the basis of targeting dual apoptotic pathways or different mechanisms of action or overcoming resistance to 1 of the agents. A list of these combinations is given in Table 2.

Finally, an attempt to use genetic data in treatment selection has been proposed on the basis of patients' myeloma cell genetic characteristics, for example, for

those patients displaying t(4;14). A few studies did show that t(4;14) patients may benefit from the use of bortezomib, either as induction therapy or as long-term treatment (56–58). In some of these studies, the long-term use of bortezomib totally overcame the poor prognosis associated with t(4;14); refs. (56, 57). For other high-risk parameters such as del(17p), or gene expression–defined high-risk disease, no specific treatment has shown beneficial effect to date. Another important question would be to define a standard of care for very good risk patients. However, these patients are not yet clearly identified, and long-term analyses are needed to try to define these patients and, then, to possibly propose less toxic approaches for these patients. Lastly, a major objective for individualized therapeutic approaches would be to define what is the best frontline or subsequent line combination for a specific patient. This objective requires genomic studies in well-defined populations of patients, treated with a specific combination (such as bortezomib-dexamethasone or lenalidomide-dexamethasone), with a primary endpoint based on progression-free survival. Several studies are currently ongoing.

## Sequencing

Recently, 2 reports have presented data on sequencing in myeloma. The first study used massively parallel whole genome-paired end sequencing, on 2 myeloma patient samples collected 6 months apart, and identified 29 somatic rearrangements, including 3 that were present only in the second sample (59). One of these rearrangements was on chromosome 13. Breakpoint sequencing revealed a 64.9-kb homozygous (no wild-type-read pairs found) deletion involving the first 2 exons of the *RB1* gene. No



**Figure 2.** High-throughput genomic analysis spanning all regulatory checkpoints. Genomic information is translated through various processes, including posttranslational protein modification (middle row). Abnormalities at these various levels potentially play a role in development of malignant transformation and behavior of the cancer cell (bottom row). Various high-throughput genomic analysis methods and arrays spanning all regulatory checkpoints are available to identify these various genomic abnormalities, to develop an integrated approach that will lead to understanding of the molecular pathogenesis of cancer, identification of novel targets and therapies, development of personalized medicine, and predictive models for outcome. \*Protein modification such as phosphorylation, acetylation, ubiquitination, sumoylation, etc.

reads spanning this breakpoint were found in the matching sample taken 6 months earlier. A second, much larger effort in 29 patients (22 whole genomes and 17 whole exomes) used 30× coverage, deep-sequencing–identified number of unique recurrent, biologically important mutations involving histone methyltransferases, transcription factor IRF4, BRAF, genes involved in protein translation, and, surprisingly, genes involved in blood coagulation (60). The results of these early sequencing efforts could provide important insights into the pathogenesis of disease progression and confirm the potential of whole genome sequencing to inform the disease biology, which may affect future therapeutic approaches.

## Conclusion

In summary, all reported studies thus far show that myeloma is characterized by a wide molecular heterogeneity. The next steps will consist in developing a combination of several molecular approaches (Fig. 2), including copy number change analyses, gene expression

profiling, massive parallel sequencing, miR analyses, and epigenetic changes surveys, in large uniformly treated patient cohorts, in order to obtain a clear landscape of the molecular changes, and their impact in myeloma classification, prognosis, and, ultimately, therapeutic management.

## Disclosure of Potential Conflicts of Interest

N.C. Munshi, consultant, Celgene, Novartis, Onyx, Millennium. H. Avet-Loiseau disclosed no potential conflicts of interest.

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